

v1 Small-Grid Dataset Report

288-Regime Surrogate Dataset Generation and QA

Codex-assisted surrogate data review

June 18, 2026

Abstract

This report reviews the Stage 2 v1 small-grid dataset generation for the American option risk-surface project. The generation used exactly the 288 approved regimes from the construction plan, the validated baseline American Crank–Nicolson / projected SOR solver, and the established LCP, boundary, and Greek diagnostic layers. It produced one compressed `.npz` package, human-audit manifests, QA summaries, QA figures, and a higher-grid confirmation table. It did not train a neural model, create `.pt` files, switch to Rannacher smoothing, or generate regimes outside the approved v1 plan. The v1 review decision is `READY_FOR_PRICE_SURROGATE_PLANNING`.

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1 Purpose of v1 Small-Grid Dataset

The v1 small-grid dataset is the first approved full small-grid dataset package after the v0 dry-run generation test. Its purpose is to create a compact, auditable dataset for future price-surrogate planning. The dataset stores normalized features, price and continuation-premium labels, diagnostic boundary and Greek fields, masks, solver metadata, split assignments, and QA manifests.

This is not a production dataset and not a model-training stage. It is a small-grid dataset and QA gate that comes before any neural surrogate experiment.

2 Roadmap Link and v0 Result

The compressed roadmap places this work in Stage 2: v1 small-grid dataset generation plus dataset QA. The preceding v0 dry run accepted all eight dry-run regimes and produced 41,624 accepted sampled rows. That result supported moving from pipeline testing to the approved 288-regime small-grid generation.

The solver source remains:

```
solver_name=american_crank_nicolson_psor_price
solver_variant=baseline_cn_psor
```

The numerical foundation follows the validated American finite-difference and LCP framework used throughout the solver ladder (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

3 Regime Count and Split Counts

The v1 generation used the exact Cartesian product approved in the construction plan:

$$\{\text{put, call}\} \times \{0.25, 0.5, 1.0, 2.0\} \times \{0.20, 0.40, 0.60\} \times \{0.01, 0.05, 0.10\} \times \{0.00, 0.03, 0.06, 0.10\}.$$

This gives 288 regimes: 144 puts and 144 calls. No-dividend call regimes remain explicit control regimes rather than dividend-call early-exercise evidence.

Split	Regimes	Accepted rows	Purpose
Train	202	1,051,006	Main fitting split for future price-surrogate planning.
Validation	19	98,857	Regime-level validation split.
Test	43	223,729	Regime-level test and control regimes.
Stress holdout	24	124,872	Held-out stress combinations.
Total	288	1,498,464	All accepted v1 sampled rows.

4 Solver and Grid Settings

Every v1 regime used $K = 1$, $S_{\max} = 4$, $M = N = 120$, and the baseline CN/PSOR solver. Sampling was restricted to the approved reporting and learning region $S/K \in [0.4, 1.8]$ and $\tau/T \in [0, 1]$. The full solver domain was retained internally for boundary conditions and diagnostics.

Each accepted regime contributed 5,203 sampled rows: 43 spot nodes by 121 time rows. The compressed package size is 50,262,900 bytes.

5 Feature, Label, and Mask Schema

The feature matrix uses:

$$x = (\log(S/K), \tau/T, r, q, \sigma, T, \text{is_call}).$$

The stored labels are:

- core labels: `value_over_K`, `payoff_over_K`, and `premium_over_K`;
- diagnostic labels: `exercise_indicator` and `boundary_spot_over_K`;
- cautioned Greek diagnostics: `delta` and `scaled_gamma`.

The mask matrix contains payoff-kink-near, boundary-near, maturity-row, strict-interior, Gamma-allowed, Delta-allowed, exercise-region, and continuation-region masks. These masks are not optional: they are the main protection against treating boundary and Gamma diagnostics as production-quality labels.

6 Diagnostics and Acceptance Checks

A regime was accepted only if PSOR converged, obstacle violation was at most 10^{-8} , equation violation was at most 10^{-6} , complementarity product was at most 10^{-6} , and required metadata and masks were present.

Metric	v1 result	Interpretation
Regimes solved	288	Exactly the approved v1 plan.
Accepted regimes	288	All regimes passed acceptance checks.
Accepted sampled rows	1,498,464	5,203 rows per accepted regime.
Maximum obstacle violation	0.0	Payoff obstacle respected in accepted regimes.
Maximum equation violation	1.02×10^{-7}	Below the 10^{-6} threshold.
Maximum complementarity product	1.82×10^{-8}	Below the 10^{-6} threshold.
Boundary-found time rows	29,795	Boundary metadata was recorded across regimes.
Strict Greek mask nodes	3,825,842	Greek diagnostics remain mask-dependent.

7 Output Files and Manifests

Output	Type	Purpose
results/04_surrogate_dataset/v1_small_grid/dataset_v1_small_grid.npz	.npz	Accepted-row v1 arrays only.
results/04_surrogate_dataset/v1_small_grid/regime_manifest.csv	CSV	Per-regime parameters, diagnostics, and acceptance status.
results/04_surrogate_dataset/v1_small_grid/diagnostic_summary.csv	CSV	PSOR, LCP, boundary, Greek, and runtime diagnostics.
results/04_surrogate_dataset/v1_small_grid/split_assignment.csv	CSV	Regime-level split assignment and row counts.
results/04_surrogate_dataset/v1_small_grid/schema_snapshot.csv	CSV	Feature, label, mask, and audit-field schema.
results/04_surrogate_dataset/v1_small_grid/output_manifest.csv	CSV	Output audit manifest for v1 artifacts.
results/04_surrogate_dataset/v1_small_grid/generation_log.txt	TXT	Plain-text generation progress log.
results/04_surrogate_dataset/v1_small_grid/v1_dataset_quality_summary.csv	CSV	Aggregate QA status and decision metrics.
results/04_surrogate_dataset/v1_small_grid/v1_higher_grid_confirmation.csv	CSV	QA-only $M = N = 180$ confirmation checks.

The compressed package contains the keys `X`, `y_value`, `y_payoff`, `y_premium`, `y_exercise_indicator`, `y_boundary`, `y_delta`, `y_scaled_gamma`, `masks`, `regime_index`, `feature_names`, `label_names`, `mask_names`, `audit_numeric`, `audit_numeric_names`, `regime_ids`, and `split_names`.

8 QA Figures

The QA figures are diagnostic only. They check row balance, diagnostic tolerances, premium distribution, boundary metadata availability, and Greek-mask availability.

9 Higher-Grid Confirmation

Four representative regimes were rerun at $M = N = 180$ for QA only. These confirmation rows were not mixed into the v1 dataset arrays. All four confirmation runs passed the same diagnostic acceptance rule. The maximum selected-node value difference versus the v1 grid was about 1.11×10^{-2} , with mean

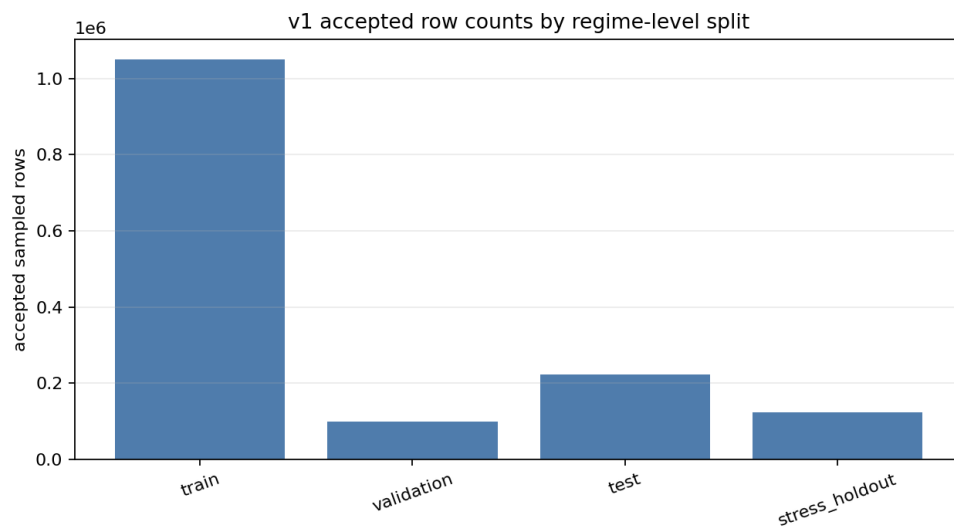


Figure 1: Accepted sampled rows by regime-level split. The counts match the deterministic construction plan.

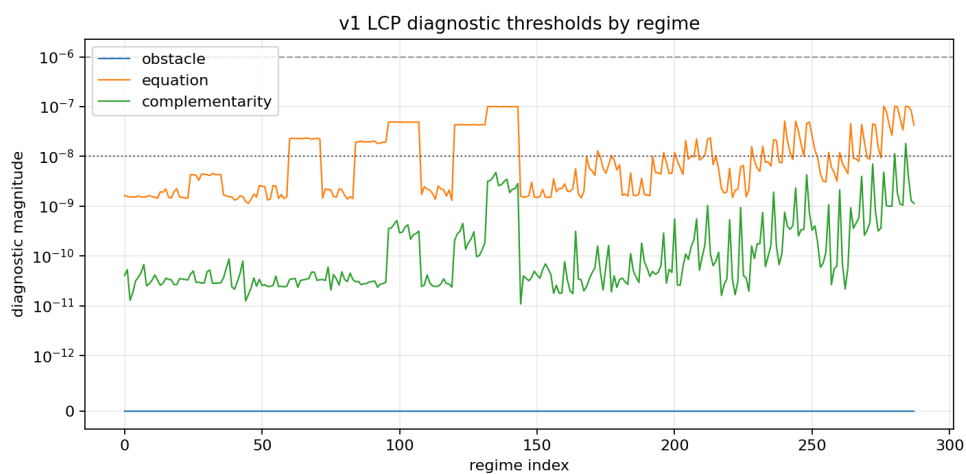


Figure 2: Obstacle, equation, and complementarity diagnostics by regime. The maxima remain below the v1 acceptance thresholds.

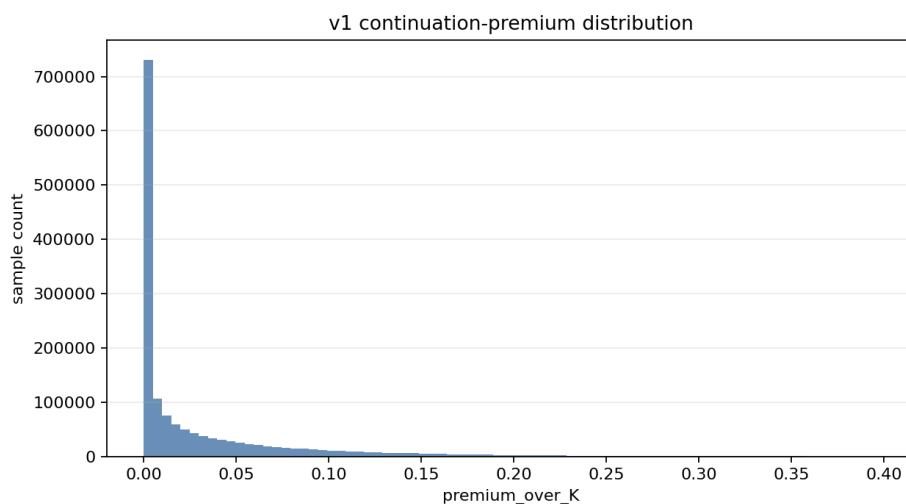


Figure 3: Continuation-premium distribution over accepted rows. This supports price and premium label QA, not production risk claims.

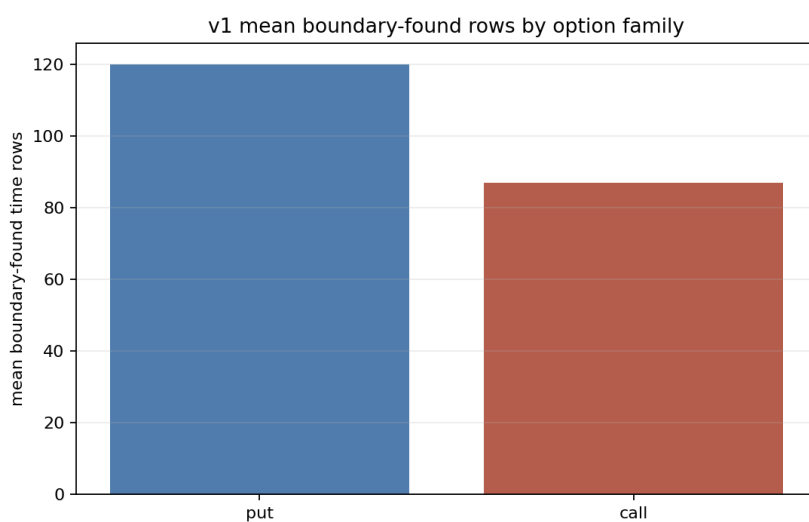


Figure 4: Mean boundary-found time rows by option family. Boundary values remain threshold-based diagnostics.

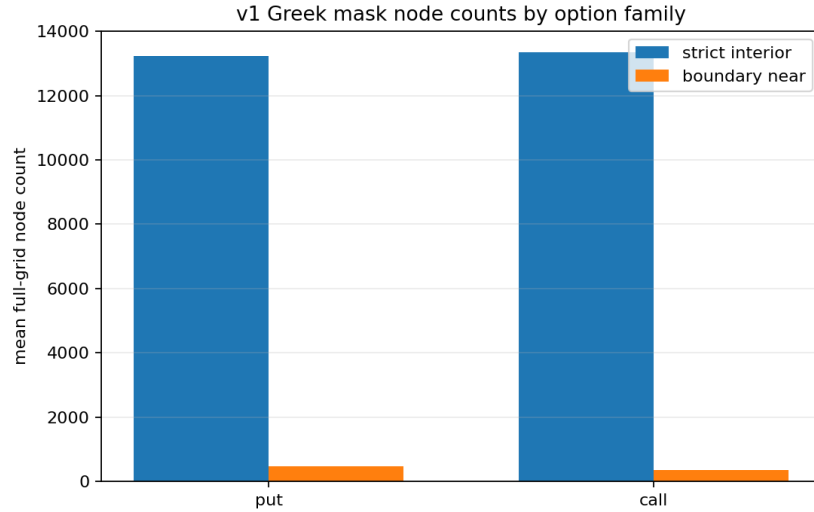


Figure 5: Greek mask node counts by option family. Gamma remains a cautioned diagnostic field and should be used only with masks.

absolute value differences near 1.98×10^{-3} . This confirms basic scale consistency while preserving the v1 dataset as an $M = N = 120$ package.

10 v1 Dataset Review Decision

The v1 review decision is:

READY_FOR_PRICE_SURROGATE_PLANNING

This means the small-grid package is ready to support a future price-surrogate planning step. It does not automatically approve neural training, production deployment, or unmasked Greek labels.

11 Limitations

- The dataset is a small-grid package, not a production-scale dataset.
- Boundary labels remain threshold-based diagnostics, not exact free-boundary truth.
- Delta and scaled Gamma remain diagnostic fields. Gamma is high caution near payoff kinks, maturity rows, and extracted boundary bands.
- The $M = N = 180$ confirmation is small and QA-oriented; it is not a full convergence proof.
- No neural model has been trained or validated from this dataset.

12 What This Dataset Supports

The v1 package supports price-surrogate planning with normalized features, reliable price and premium labels, split metadata, diagnostic boundary fields, Greek masks, solver diagnostics, and regime-level

manifests. It also supports human review of whether the first surrogate experiment should target price and continuation premium before using any Greek diagnostic targets.

13 What This Dataset Does Not Support

The v1 package does not support production risk claims, final paper conclusions by itself, production Greek labels, broad extrapolation to regimes outside the approved grid, or automatic approval to train neural surrogates without a separate surrogate-planning gate.

14 Recommended Next Step

The recommended next step is a price-surrogate planning gate. That gate should decide the first model family, training objective, loss functions, regime-level evaluation protocol, and whether Greek fields remain metadata-only for the first surrogate experiment.

References

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